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**MACHINE LEARNING APPROACHES FOR  
PREDICTING TRANSLATION EFFICIENCY  
FROM CODING SEQUENCES**

Helena Floreani

Zagreb, June, 2026.

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**PRISTUPI STROJNOG UČENJA ZA  
PREDVIĐANJE UČINKOVITOSTI TRANSLACIJE  
IZ KODIRAJUĆIH SEKVENCI**

Helena Floreani

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## BACHELOR THESIS ASSIGNMENT No. 2267

Student: **Helena Floreani (0036559207)**

Study: Electrical Engineering and Information Technology and Computing

Module: Computing

Mentor: prof. Mile Šikić, PhD

Title: **Machine Learning Approaches for Predicting Translation Efficiency from Coding Sequences**

### Description:

Ribonucleic acid (RNA) is a central biomolecule involved in gene expression and regulation. In particular, messenger RNA (mRNA) encodes proteins, and its sequence composition influences properties such as stability and translation efficiency. Understanding how sequence features affect protein output is especially important for the design of RNA-based therapeutics. Recent advances in machine learning, especially Transformer-based language models, have shown that neural networks trained on large-scale sequence data can learn rich representations of complex patterns. Originally developed for natural language processing, these approaches are increasingly applied to biological sequences, where nucleotides and codons can be treated analogously to words. The aim of this project is to investigate how well translation efficiency can be predicted from coding sequence (CDS) using machine learning methods. Predictive models will be developed using sequence-derived features, and pretrained RNA language models will be integrated into the predictive framework to assess whether learned sequence representations improve performance. Different modeling strategies will be systematically evaluated to determine their ability to capture biologically relevant signals. The project will involve feature extraction from coding sequences, implementation of predictive models in Python, and systematic evaluation using an existing dataset. The code should be well-structured, reproducible, and follow standard Python programming practices.

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## **ZAVRŠNI ZADATAK br. 2267**

Pristupnica: **Helena Floreani (0036559207)**  
Studij: Elektrotehnika i informacijska tehnologija i Računarstvo  
Modul: Računarstvo  
Mentor: prof. dr. sc. Mile Šikić

Zadatak: **Pristupi strojnog učenja za predviđanje učinkovitosti translacije iz kodirajućih sekvenci**

### Opis zadatka:

Ribonukleinska kiselina (RNA) središnja je biomolekula uključena u ekspresiju i regulaciju gena. Posebno, glasnička RNA (mRNA) kodira proteine, a sastav njezine sekvence utječe na svojstva poput stabilnosti i učinkovitosti translacije. Razumijevanje kako značajke sekvence utječu na količinu proizvedenog proteina izuzetno je važno za razvoj terapeutika temeljenih na RNA. Nedavni napredak u strojnome učenju, osobito jezičnih modela temeljenima na Transformer arhitekturi, pokazao je da neuronske mreže trenirane na velikim skupovima sekvencijskih podataka mogu naučiti bogate reprezentacije složenih obrazaca. Izvorno razvijeni za obradu prirodnog jezika, ovi se pristupi sve češće primjenjuju na biološke sekvence, gdje se nukleotidi i kodoni mogu promatrati analogno riječima. Cilj ovog projekta je istražiti koliko se učinkovito učinkovitost translacije može predvidjeti iz kodirajuće sekvence (CDS) primjenom metoda strojnog učenja. Razvit će se prediktivni modeli temeljeni na značajkama izvedenima iz sekvence, a u sustav će se uključiti i predtrenirani RNA jezični modeli kako bi se procijenilo poboljšavaju li naučene reprezentacije rezultate. Različite strategije modeliranja sustavno će se vrednovati s obzirom na sposobnost prepoznavanja biološki relevantnih signala. Projekt uključuje ekstrakciju značajki iz kodirajućih sekvenci, implementaciju modela u Pythonu te sustavnu evaluaciju na postojećem skupu podataka. Kod treba biti strukturiran, reproduktibilan i u skladu sa standardnim praksama programiranja u Pythonu.

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*I would like to sincerely thank my advisor, Prof. Mile Šikić, Ph.D., for his mentorship, guidance, and support throughout the course of this thesis. Special thanks go to Ivona Martinović, mag. ing. comp., for her continuous assistance, valuable advice, and patience during every stage of this work.*

# **Abstract**

## **Machine Learning Approaches for Predicting Translation Efficiency from Coding Sequences**

Helena Floreani

Ribonucleic acid (RNA) plays an essential role in gene expression. One important type of RNA, messenger RNA (mRNA), serves as the template for protein synthesis. Translation efficiency measures how well an mRNA transcript is translated into protein and is influenced by a variety of factors, including sequence characteristics, transcript-specific regulatory elements, and the cellular environment. Predicting translation efficiency accurately helps us understand gene regulation and supports the development of RNA-based therapeutics.

This thesis explores how to predict translation efficiency from mRNA sequence data using machine learning. First, a dataset of human endogenous transcripts was analyzed to examine translation efficiency values and biologically relevant sequence characteristics. Several biologically motivated sequence-derived features were extracted from the transcripts, including nucleotide composition, codon usage, Codon Adaptation Index, GC content, GC3 content, and sequence length.

Traditional machine learning models, including linear regression and LightGBM, were trained using biologically motivated sequence-derived features. In addition, embeddings generated by a pretrained RNA language model were used as input features for prediction. Finally, the pretrained RNA language model was directly adapted to the translation efficiency prediction task through fine-tuning. The performance of all approaches was evaluated and compared.

The handcrafted feature-based LightGBM model had the best performance among the models we developed in this study ( $R^2 = 0.504$ ). The top embedding-based model reached an  $R^2$  of 0.432. Still, neither model matched the RiboNN baseline, which achieved  $R^2 = 0.584$  for LightGBM and  $R^2 = 0.600$  for the deep neural network on the same test set. The fine-tuned RiNALMo model performed much worse.

Overall, these results show that carefully designed sequence-derived features remain highly competitive for translation efficiency prediction. At the same time, the strong performance of the RiboNN deep neural network demonstrates that specialized deep learning models trained directly on sequence data can provide additional predictive value. In the evaluated experimental setup, RiNALMo embeddings and the fine-tuned RiNALMo model did not match the performance of the feature-based approaches. Nevertheless, pretrained RNA language models remain a promising research direction, and future work should investigate alternative pretrained models and more extensive fine-tuning strategies.

**Keywords:** RNA, messenger RNA, translation efficiency, machine learning, Transformer-based architecture, RNA language model, fine-tuning

# Sažetak

## **Pristupi strojnog učenja za predviđanje učinkovitosti translacije iz kodirajućih sekvenci**

Helena Floreani

Ribonukleinska kiselina (RNA) ima ključnu ulogu u ekspresiji gena. Jedna od najvažnijih vrsta RNA je glasnička RNA (mRNA), koja služi kao predložak za sintezu proteina. Translacijska učinkovitost opisuje koliko se učinkovito mRNA transkript prevodi u protein te ovisi o različitim značajkama sekvence. Točno predviđanje translacijske učinkovitosti važno je za razumijevanje regulacije gena i razvoj terapija temeljenih na RNA.

U ovom radu istražuje se predviđanje translacijske učinkovitosti na temelju podataka o mRNA sekvencama primjenom metoda strojnog učenja. Najprije je provedena analiza skupa podataka kako bi se ispitale vrijednosti translacijske učinkovitosti i obilježja sekvenci. Iz transkripata su izdvojena biološki motivirana obilježja, uključujući sastav nukleotida, uporabu kodona, indeks prilagodbe kodona (CAI), GC sadržaj, GC3 sadržaj i obilježja duljine sekvence.

Tradicionalni modeli strojnog učenja, uključujući linearnu regresiju i LightGBM, trenirani su na temelju biološki motiviranih obilježja izvedenih iz sekvenci. Osim toga, kao ulazna obilježja za predviđanje korištene su reprezentacije dobivene prednaučenim RNA jezičnim modelom. Naposljetku je prednaučeni RNA jezični model dodatno prilagođen zadatku predviđanja translacijske učinkovitosti postupkom finog podešavanja. Uspješnost svih pristupa procijenjena je i međusobno uspoređena.

Model LightGBM temeljen na ručno definiranim obilježjima ostvario je najbolje rezultate među modelima razvijenima u ovom radu ( $R^2 = 0.504$ ). Najuspješniji model



temeljen na reprezentacijama ostvario je vrijednost  $R^2 = 0.432$ . Međutim, niti jedan od tih modela nije dosegao uspješnost referentnih modela RiboNN, koji su na istom testnom skupu ostvarili  $R^2 = 0.584$  za LightGBM i  $R^2 = 0.600$  za model dubokog učenja. Fino podešeni model RiNALMo ostvario je znatno slabiju prediktivnu uspješnost.

Općenito, dobiveni rezultati pokazuju da pažljivo definirana obilježja izvedena iz sekvenci i dalje predstavljaju vrlo konkurentan pristup predviđanju translacijske učinkovitosti. Istodobno, visoka uspješnost modela dubokog učenja RiboNN pokazuje da specijalizirani modeli dubokog učenja, trenirani izravno na podacima o sekvencama, mogu pružiti dodatnu prediktivnu vrijednost. U eksperimentalnom postavu primijenjenom u ovom radu, reprezentacije modela RiNALMo i njegovo fino podešavanje nisu ostvarili uspješnost usporedivu s pristupima temeljenima na ručno definiranim obilježjima. Unatoč tome, prednaučeni RNA jezični modeli i dalje predstavljaju obećavajući smjer budućih istraživanja, pri čemu bi budući radovi trebali ispitati druge prednaučene modele i opsežnije strategije finog podešavanja.

**Ključne riječi:** RNA, glasnička RNA, učinkovitost translacije, strojno učenje, Transformer arhitektura, RNA jezični model, fino podešavanje

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# 1 Introduction

Ribonucleic acid (RNA) plays a fundamental role in gene expression and regulation. In particular, messenger RNA (mRNA) serves as the template for protein synthesis by carrying genetic information from DNA to the cellular machinery responsible for translation [1, 2]. Translation efficiency describes how effectively an mRNA transcript is translated into protein. It is influenced by various sequence-dependent factors and represents an important aspect of post-transcriptional gene regulation [3, 4].

Accurate prediction of translation efficiency has attracted considerable interest due to its relevance in understanding gene regulation and developing RNA-based therapeutics [5]. In applications such as mRNA vaccines, therapeutic effectiveness depends on the amount of protein produced from the delivered mRNA sequence, making translation efficiency an important design consideration. Traditional prediction approaches often rely on biologically motivated sequence-derived features, such as GC content and codon usage statistics. Although informative, these handcrafted features may not fully capture the complex sequence patterns that influence protein production.

Recent advances in machine learning, particularly the introduction of the Transformer architecture by Vaswani et al. [6], have enabled the development of powerful language models capable of learning informative representations directly from large collections of biological sequences. Inspired by advances in natural language processing, RNA language models treat nucleotide sequences as a form of biological language and learn sequence representations that capture meaningful structural and functional properties. Examples of such models include RNA-FM [7], RiNALMo [8], DNABERT [9] and ProtBERT [10].

The objective of this work is to study machine learning approaches for predicting

translation efficiency from RNA sequences. First, an exploratory analysis of the translation efficiency dataset containing human endogenous transcripts, from the RiboNN framework [11], is performed. The analysis includes the examination of sequence characteristics and biologically motivated sequence-derived features such as GC content, GC3 content, and the Codon Adaptation Index (CAI). These features are subsequently used as inputs to predictive models. In addition, the predictive value of learned sequence representations generated by the pretrained RiNALMo RNA language model [8] is evaluated and compared with handcrafted sequence-derived features. The potential application of fine-tuning pretrained RNA language models for translation efficiency prediction is also considered.

The remainder of this thesis is organized as follows. Chapter 2 presents the theoretical background related to RNA biology, translation efficiency, machine learning methods used in this thesis, and RNA language models. Chapter 3 presents the methodology of this study, covering the dataset, feature extraction procedures, generation of RiNALMo-based representations, predictive models, and evaluation metrics used to assess translation efficiency prediction performance. Chapter 4 presents and discusses the experimental results obtained using sequence-derived features, pretrained RNA language model representations, and model fine-tuning. Finally, Chapter 5 summarizes the study's main findings, discusses its limitations, and outlines directions for future research.

## 2 Theoretical background

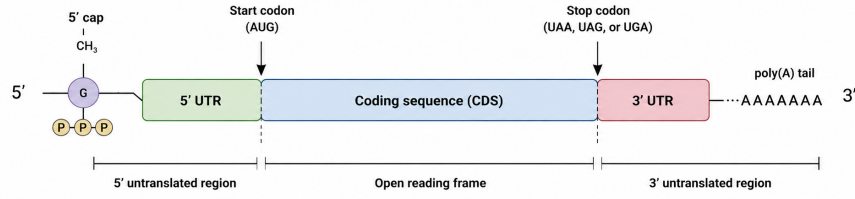
This chapter provides the biological and computational background required to understand translation efficiency prediction. Section 2.1 introduces the process of mRNA translation and the concept of translation efficiency. Section 2.2 presents the RiboNN framework [11] and the dataset used in this work. Handcrafted features and machine learning approaches for biological sequence analysis are described in Sections 2.3 and 2.4, respectively. Finally, Sections 2.5 and 2.6 introduce Transformer-based language models and the RiNALMo RNA language model [8], which form the basis for the representation-learning approaches investigated in this study.

### 2.1 Translation and translation efficiency

Ribonucleic acid (RNA) plays a key role in gene expression by carrying and processing genetic information inside the cell [1, 2]. RNA is made up of four nucleotides: adenine (A), uracil (U), cytosine (C), and guanine (G). Messenger RNA (mRNA) is a type of RNA that serves as an intermediary between DNA and protein synthesis by carrying genetic information to ribosomes [2], whereas non-coding RNAs (ncRNAs) participate in various regulatory and structural cellular processes.

A typical eukaryotic mRNA transcript has several parts: a 5' cap, a 5' untranslated region (5' UTR), a coding sequence (CDS), a 3' untranslated region (3' UTR), and a poly(A) tail [4]. These parts affect how the mRNA functions, including its stability, where it goes in the cell, and how it is translated. Figure 2.1 shows the organization of a mRNA molecule and the relative positions of the 5'UTR, CDS and 3'UTR regions.

During translation, ribosomes read the mRNA's coding sequence in sets of three nucleotides, called codons. Each codon specifies a particular amino acid that is added to



**Figure 2.1:** General structure of a eukaryotic mRNA transcript showing the 5' cap, 5' UTR, coding sequence (CDS), 3' UTR, and poly (A) tail.

the growing protein chain [2].

The genetic code is degenerate, meaning that most amino acids can be encoded by multiple codons. As a result, multiple mRNA sequences can encode the same protein even though they have different nucleotide sequences. These synonymous codons are not used with equal frequency, and their usage has been shown to influence translation efficiency and other aspects of gene expression [12, 3].

Translation efficiency (TE) quantitatively describes the effectiveness with which an mRNA transcript is translated into protein. Multiple factors influence TE, such as codon usage, nucleotide composition, untranslated regions, RNA secondary structure, and transcript stability [3, 4].

TE is an important part of post-transcriptional gene regulation. Understanding what affects translation efficiency is key for basic biology and for developing RNA-based therapeutics. As a result, much research has aimed to identify sequence features linked to TE and to build computational models that predict translation efficiency from RNA sequence data [3].

To predict translation efficiency, researchers need TE values measured experimentally, typically using high-throughput sequencing methods [11]. The next section explains the RiboNN framework and gives details about the dataset used in this study.

## 2.2 RiboNN and translation efficiency datasets

Translation efficiency is influenced by many factors beyond RNA sequence alone, making accurate prediction a challenging task. Instead, it is typically estimated using experimental approaches that quantify both transcript abundance and translational activity. Today, most studies use a combination of RNA sequencing (RNA-seq) and ribosome profiling (Ribo-seq) to estimate translational output for thousands of transcripts at once [13].

One recent framework developed for translation efficiency prediction is RiboNN [11]. RiboNN is a deep learning model that predicts translation efficiency directly from mRNA sequence information. The framework was trained using large-scale experimental measurements obtained from multiple cell types and integrates information from different transcript regions to model factors influencing protein production.

In addition to proposing a predictive model, the RiboNN framework provides a publicly available dataset containing translation efficiency estimates and corresponding RNA sequences. This dataset serves as the basis for the analyses in this thesis and enables investigation of both traditional sequence-derived features and learned sequence representations.

Since translation efficiency is influenced by multiple sequence-dependent characteristics, a variety of biologically motivated features have been proposed to capture information relevant for prediction tasks. Several of these commonly used sequence-derived features are presented in the following section.

## 2.3 Handcrafted features

Many properties of mRNA sequences affect translation efficiency. To explore how sequence composition relates to translation, researchers often extract features directly from coding sequences. Commonly used sequence-derived features include GC content, GC3 content, the Codon Adaptation Index (CAI), sequence length measures, codon usage statistics, and nucleotide composition features. Among these, GC content, GC3 content, and CAI are particularly well-established biologically motivated descriptors and are therefore described in more detail below [3]. These features reflect different aspects of nucleotide composition and codon usage. They are widely used in gene expression stud-



ies and in predicting translation efficiency, so they are included in this work [3, 11, 14].

### 2.3.1 GC content

GC content is the proportion of guanine (G) and cytosine (C) nucleotides in a coding sequence. It is a widely used feature in genomics because it affects the physical and structural properties of nucleic acids [3].

GC content is calculated using the following formula:

$$GC = \frac{G + C}{A + U + G + C} \quad (2.1)$$

where  $A$ ,  $U$ ,  $G$ , and  $C$  represent the number of adenine, uracil, guanine, and cytosine nucleotides, respectively.

Higher GC content sequences are typically more thermodynamically stable. This is due to the fact that A-U base pairings only form two hydrogen bonds, while G-C base pairs form three [14]. The additional hydrogen bond increases the stability of local RNA structures and generally makes them more resistant to unfolding.

Studies have shown that nucleotide composition helps predict translation efficiency [15]. It is often included as a feature in machine learning models for gene expression analysis [11].

### 2.3.2 GC3 content

GC3 content is the proportion of guanine and cytosine nucleotides found at the third position of codons in a coding sequence. Since many amino acids are encoded by more than one codon, changes at this third position often do not change the amino acid. This makes the third position especially useful for studying codon usage and translation regulation [?]. GC3 content is defined as

$$GC3 = \frac{G_3 + C_3}{N_3} \quad (2.2)$$

Here,  $G_3$  and  $C_3$  are the numbers of guanine and cytosine nucleotides at the third codon position, and  $N_3$  is the total number of codons in the coding sequence.

Differences in GC3 content have been linked to changes in gene expression and translation efficiency [3]. For this reason, GC3 content is an important feature for studying coding sequences and factors that affect protein production.

### 2.3.3 Codon Adaptation Index

The Codon Adaptation Index (CAI) is a common measure of codon usage bias [12]. It quantifies the extent to which a coding sequence uses codons that are often found in highly expressed genes of an organism.

Many amino acids are encoded by several synonymous codons, but these codons are not used the same in an organism. Codons that appear more in highly expressed genes are called optimal codons, as they are considered to be more efficient in translation [12, 3].

CAI is calculated as the geometric mean of codon weights across a coding sequence:

$$CAI = \left( \prod_{i=1}^L w_i \right)^{1/L} \quad (2.3)$$

Here,  $L$  is the number of codons in the CDS, and  $w_i$  is the relative adaptiveness weight of the  $i$ -th codon. The weight is calculated as

$$w_i = \frac{f_i}{f_{\max}}, \quad (2.4)$$

where  $f_i$  is the frequency of the  $i$ -th codon and  $f_{\max}$  is the frequency of the most frequently used synonymous codon encoding the same amino acid. As a result, the most frequently used synonymous codon has a weight of 1, whereas less frequently used synonymous codons receive values between 0 and 1.

CAI values range from 0 to 1. Higher values indicate a stronger preference for optimal codons and are often associated with higher translation efficiency and protein ex-

pression. Since codon usage affects translation and ribosome activity, CAI is often used to predict gene expression and protein production. For this reason, CAI is one of the sequence-derived features studied in this work.

Together, GC content, GC3 content, and CAI represent biologically motivated sequence-derived features that summarize different aspects of nucleotide composition and codon usage.

## **2.4 Machine learning approaches for translation efficiency prediction**

Machine learning is now widely used to analyze biological sequences and predict molecular properties using experimental data [11, 3, 9, 8]. For translation efficiency prediction, these models are trained to identify relationships between sequence features and measured translation efficiency values.

Common machine learning methods include linear models, random forests, gradient boosting, and neural networks. Among feature-based machine learning approaches, gradient boosting methods have become particularly popular due to their strong predictive performance on structured data. LightGBM (Light Gradient Boosting Machine) [16] is a gradient boosting framework that builds an ensemble of decision trees in a sequential manner, where each new tree aims to correct the errors of previously constructed trees. Owing to its efficiency and predictive accuracy, LightGBM has been successfully applied to a variety of biological prediction tasks, including translation efficiency prediction [11].

One key benefit of feature-based methods is their ease of interpretation, as each feature typically has a clear biological meaning. However, how well these methods work depends on how good and complete the chosen features are. Some important sequence patterns that affect translation efficiency might not be captured by a fixed set of hand-crafted features.

The RiboNN framework explored deep learning approaches specifically designed for translation efficiency prediction. It combines sequence-derived features with a custom neural network architecture trained directly for this task [11]. While such models can

capture complex relationships among input features, they are designed specifically for translation-efficiency prediction and do not learn general-purpose RNA sequence representations.

To overcome these limitations, recent studies have explored representation-learning approaches that directly learn sequence representations from large-scale biological data [8]. Many modern RNA language models are based on the Transformer architecture [6], which enables them to capture dependencies and relationships within biological sequences.

## **2.5 Transformer architecture**

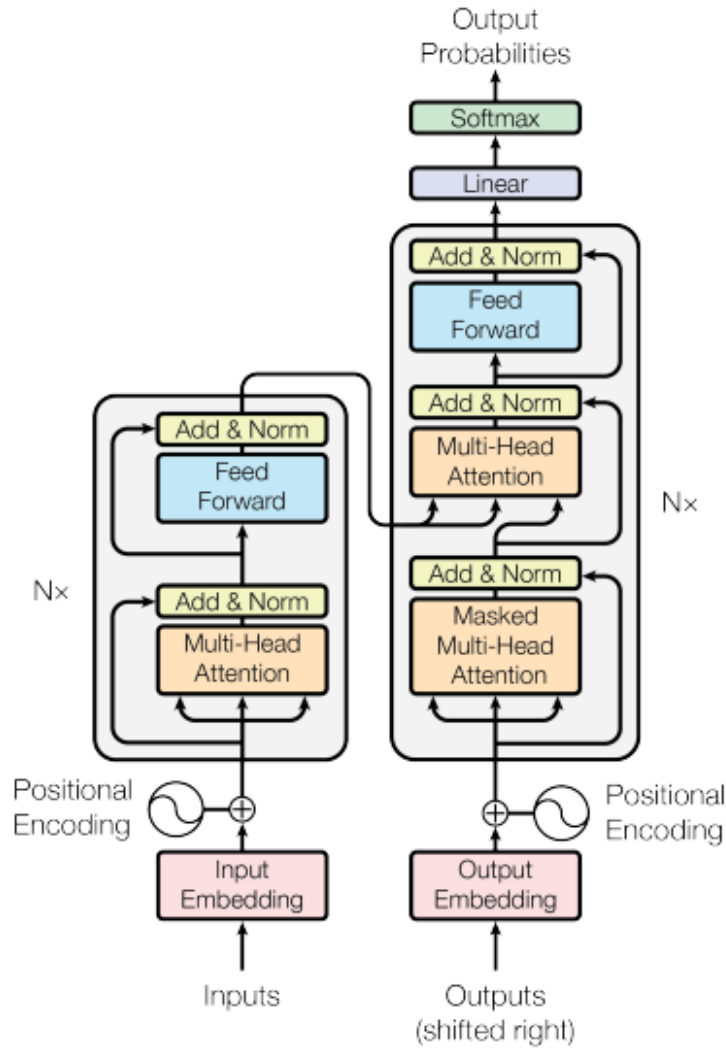
The Transformer architecture, first introduced by Vaswani et al [6], is now one of the most important deep learning models for sequence analysis. While Transformers were originally designed for natural language processing, they have also performed well in other fields, such as bioinformatics. Because they can handle sequential data and learn useful patterns from large datasets, Transformers have enabled the development of biological language models for DNA, RNA, and protein sequences [9, 7, 10].

Transformer-based models can be trained on large sets of unlabeled sequences using self-supervised learning objectives. As a result, they can learn representations that capture meaningful biological information and can subsequently be applied to downstream tasks, such as translation efficiency prediction. Understanding how these representations are learned requires examining the underlying Transformer architecture.

### **2.5.1 Encoder and decoder architectures**

Figure 2.2 shows that the original Transformer architecture has two main parts: an encoder and a decoder. The encoder processes the input sequence to create contextual representations. The decoder uses these representations to generate an output sequence [6].

Modern Transformer models may use only the encoder, only the decoder, or both, depending on the task. Encoder-based models are often used for representation learning to create useful embeddings of input sequences. Decoder-based models are usually



**Figure 2.2:** Original Transformer architecture proposed by Vaswani et al. [6].

chosen for tasks that involve generating sequences [6].

Many biological language models, such as RNA language models, use an encoder-based architecture [8, 9]. Their main goal is to learn informative representations of biological sequences rather than create new ones. These representations, called embeddings, can be used as features for downstream prediction tasks. In addition, the pre-trained model can be fine-tuned on task-specific data to further improve its representations and adapt them to a particular application.

All encoder-based language models are pre-trained using self-supervised learning objectives, which enable the model to learn from large collections of unlabeled sequences. One commonly used objective is Masked Language Modeling (MLM), where selected se-

sequence positions are masked and the model is trained to predict the original nucleotides from their surrounding context. This approach allows the model to learn informative sequence representations that can subsequently be transferred to downstream prediction tasks.

Among RNA language models based on this paradigm, RiNALMo [8] represents one of the largest publicly available models and is the foundation of the representation learning approach investigated in this work.

## 2.6 RiNALMo

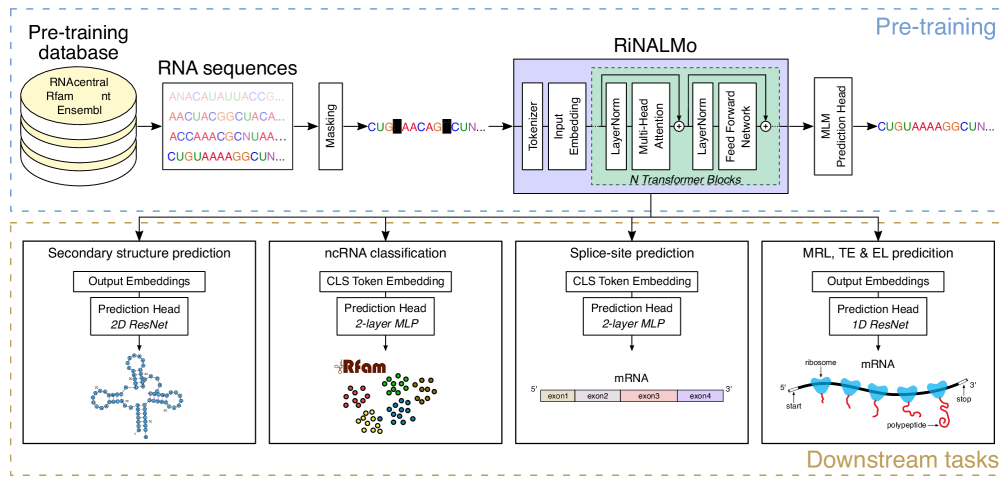
RiNALMo (RiboNucleic Acid Language Model) is a Transformer-based RNA language model designed to learn useful representations of RNA sequences [8]. It was pre-trained on about 36 million non-coding RNA sequences from several public databases and has around 650 million parameters.

Depending on the application, RiNALMo can be used either as a fixed feature extractor that generates sequence embeddings or as a pretrained model that is subsequently fine-tuned for a specific downstream prediction task.

The authors showed that RiNALMo works well on several RNA-related prediction tasks, such as secondary structure prediction, splice-site prediction, mean ribosome loading prediction, and ncRNA functional classification [8]. Because the model can generate informative sequence embeddings, it is especially useful for applications where hand-crafted sequence features might miss complex patterns.

In this study, we use embeddings from the pretrained RiNALMo model in the translation efficiency prediction pipeline and compare them with traditional sequence-derived features such as GC content, GC3 content, and the Codon Adaptation Index, while also evaluating a fine-tuned version of RiNALMo for translation efficiency prediction.

Figure 2.3 illustrates the pre-training procedure of RiNALMo and several downstream tasks for which the learned RNA representations can be applied.



**Figure 2.3:** Overview of the RiNALMo framework. Adapted from [8].

## 3 Methodology

This chapter explains the dataset, the extraction of handcrafted sequence-derived features and RiNALMo-based sequence representations, the machine learning models used, and how the results were evaluated in this study.

### 3.1 Dataset

The analyses performed in this work are based on a publicly available dataset derived from the RiboNN framework [11]. The dataset contains human endogenous RNA transcript sequences together with experimentally derived translation efficiency (TE) measurements collected across multiple cell types.

Each transcript is described by a set of attributes that include transcript identifiers, sequence lengths, nucleotide sequences, and translation efficiency values. In total, the dataset contains 10348 transcripts and 88 attributes. Translation efficiency measurements are available for 78 different cell types. Table 3.1 summarizes the main attributes contained in the dataset.

In addition to cell-type-specific translation efficiency measurements, the dataset provides the `mean_TE` attribute, which represents the average translation efficiency across all available cell types for a given transcript.

In this study, `mean_TE` was used as the target variable for predictive modeling. The nucleotide sequences contained in the dataset were further processed to extract biologically motivated sequence-derived features and to generate learned sequence representations using the pretrained RiNALMo model.



Table 3.1: Main attributes contained in the RiboNN-derived dataset.

| Attribute   | Description                                     |
|-------------|-------------------------------------------------|
| Symbol      | Gene symbol                                     |
| Gene_id     | Gene identifier                                 |
| Tx_id       | Transcript identifier                           |
| Tx_size     | Transcript length                               |
| Utr5_size   | Length of the 5' untranslated region            |
| Utr3_size   | Length of the 3' untranslated region            |
| Cds_size    | Length of the coding sequence                   |
| Tx_sequence | Nucleotide sequence of the transcript           |
| TE values   | Translation efficiency values for 78 cell types |
| Fold        | Dataset partition indicator                     |
| Mean_te     | Mean translation efficiency across cell types   |

## 3.2 Feature Extraction

Sequence-derived features were extracted from the transcripts in the RiboNN dataset to capture different aspects of sequence composition, codon usage, and transcript structure that may influence translation efficiency. These features were selected to reflect different aspects of sequence composition, codon usage, and transcript structure that could affect translation efficiency.

The dataset includes complete transcript sequences and annotations for the lengths of the 5' untranslated region (5'UTR), coding sequence (CDS), and 3' untranslated region (3'UTR). Using these annotations, we reconstructed the CDS regions from the transcript sequences and extracted features related to the coding sequence. Even though RNA sequences have uracil (U) instead of thymine (T), we kept the thymine notation because both the dataset and the feature extraction process use DNA nucleotide notation.

The features we extracted included GC content, GC3 content, and the Codon Adaptation Index (CAI), as described in Section 2.3. We also included sequence length features representing transcript structure, to be exact the lengths of the 5'UTR, coding sequence (CDS), 3'UTR, and the complete transcript. To capture codon usage, we calculated relative codon frequencies for each coding sequence. We further measured nucleotide composition by computing the proportions of adenine (A), cytosine (C), guanine (G), and uracil (U) in the CDS, 5'UTR, and 3'UTR regions.

The extracted features were combined into a feature vector, which then served as

input to the feature-based machine learning models. Later, we compared these handcrafted features with sequence representations generated by the pretrained RiNALMo language model.

### 3.3 RiNALMo based representations

Along with handcrafted sequence features, this study also looks at learned sequence representations from the pretrained RiNALMo [8]. Traditional features like GC content, GC3 content, and CAI focus on specific biological properties, but RiNALMo representations are learned from large sets of RNA sequences during pre-training.

These learned representations can capture complex sequence patterns, context, and important biological information that handcrafted features may overlook. To apply RiNALMo to predict translation efficiency, we used two approaches: using pretrained sequence embeddings and fine-tuning the pretrained model for this prediction task.

#### 3.3.1 Embedding extraction

We first used the pretrained RiNALMo model as a fixed feature extractor. The transcript sequences were tokenized using the RiNALMo tokenizer and subsequently processed by the pretrained model to obtain sequence representations.

We generated embeddings using the `multimolecule/rinalmo-giga` checkpoint [8]. For each input sequence, hidden representations from the final Transformer layer were extracted. To obtain a fixed-length representation of each transcript, the embedding corresponding to the CLS token was used as the sequence-level embedding.

These embeddings provide dense numerical representations of RNA sequences and serve as input features for downstream machine learning models. We then compared their predictive performance to models that used traditional sequence-derived features.

### 3.4 Predictive models

The handcrafted features described in Section 2.3 were used as input variables for machine learning models. In addition, embeddings generated by the pretrained RiNALMo

model were evaluated as an alternative representation for sequences. This section describes the predictive models used and the evaluation procedure applied throughout the study.

To make sure the comparison was fair, all machine learning models were trained and evaluated using the same data split. The dataset was randomly divided into training and test sets in an 80/20 ratio, with a fixed random seed (`seed = 42`). In addition, 10% of the training set was reserved as a validation set for neural network models that required it during training. Linear Regression, LightGBM, the RiboNN models, and the RiNALMo embedding-based models were therefore evaluated on the same held-out test set. Due to computational constraints, the fine-tuned RiNALMo model was trained and evaluated using the corresponding subset of transcripts shorter than 4000 nucleotides.

### **3.4.1 Linear regression**

Linear regression was used as a baseline model for predicting translation efficiency. Depending on the experiment, the model was trained either on handcrafted features or on embeddings generated by the pretrained RiNALMo model. The model assumes a linear relationship between the input features and the target variable, providing a simple and interpretable benchmark for evaluating the predictive value of learned RNA embeddings.

A linear regression model implemented in scikit-learn was trained on the training data and evaluated on the test set for both input representations.

### **3.4.2 LightGBM**

LightGBM was tested as a non-linear machine learning model to predict translation efficiency. It is a gradient boosting framework that builds decision trees one after another, with each new tree trying to fix the errors made by the previous ones. Gradient boosting decision trees are popular because they offer strong predictive power and some interpretability [16].

The LightGBM model was tested on both handcrafted features and RiNALMo embeddings as inputs. These two approaches were later compared. The `mean_TE` attribute served as the prediction target.

A LightGBM regressor was set up with the default settings from the LightGBM library. After training, predictions were made for the test set. Feature importance scores were also examined to identify which handcrafted features had the greatest impact on the prediction of translation efficiency.

### **3.4.3 RiNALMo fine-tuning**

In addition to using RiNALMo as a fixed feature extractor, the pretrained RiNALMo-Giga model was also adapted to predict translation efficiency.

A pretrained language model cannot directly predict translation efficiency. Therefore, a task-specific prediction head was added to the pretrained encoder to transform the learned sequence representations into a translation efficiency prediction. Following the methodology of the original RiNALMo study, the prediction head architecture proposed in the UTR-LM model was used [8, 17].

The prediction head has six one-dimensional residual convolutional (ResNet) blocks. Each block includes two one-dimensional convolutional layers, instance normalization, and ELU activation functions. Residual connections help with optimization and let the model capture local sequence patterns important for predicting translation efficiency.

The prediction head is a lightweight, task-specific part that turns the contextual RNA representations from the pretrained encoder into translation efficiency predictions. The pretrained RiNALMo encoder has about 650 million parameters, while the prediction head has only around 77900 trainable parameters.

Several training configurations, including different learning rates, batch sizes, numbers of training epochs, and fine-tuning schedules, were evaluated during preliminary experiments. The configuration providing the best validation performance was selected for the final model.

During the first training epoch, only the prediction head was trained while the pretrained RiNALMo encoder remained frozen. Starting from the second epoch, gradual unfreezing was applied by fine-tuning the encoder’s final-layer normalization and the last two transformer blocks, while the remaining encoder parameters remained frozen.

The model was trained for 15 epochs. Before training, transcripts longer than 4000 nucleotides were excluded from the dataset to reduce GPU memory requirements. Long sequences are particularly demanding for Transformer-based models because the memory and computational cost increase approximately quadratically with sequence length. The model was trained using the Adam optimizer and Huber loss, following the training strategy used for translation efficiency prediction in the original RiNALMo study [8]. The learning rate was initialized to  $1 \times 10^{-4}$  and linearly decayed to  $1 \times 10^{-5}$  during the first 600 optimization steps, after which it remained constant. The mini-batch size was set to 4 sequences, and gradients were accumulated over 4 mini-batches, resulting in an effective batch size of 16. All fine-tuning experiments were conducted on a single NVIDIA RTX A6000 GPU equipped with 48 GB of memory.

## 3.5 Evaluation metrics

We evaluated the performance of the predictive models using several standard regression metrics. Each metric gives different insights into prediction accuracy, how well the model explains the data, and how closely predictions match observed translation efficiency values.

### 3.5.1 Mean squared error

Mean Squared Error (MSE) shows the average squared difference between predicted and observed values [18]. It is calculated as

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad (3.1)$$

Here,  $y_i$  is the observed translation efficiency value,  $\hat{y}_i$  is the predicted value, and  $n$  is the number of samples.

The MSE is always non-negative, with 0 indicating perfect predictions. Lower MSE values, therefore, correspond to smaller prediction errors and better model performance.

### 3.5.2 Coefficient of determination

The coefficient of determination ( $R^2$ ) shows how much of the variation in the target variable the model can explain [19]. It is calculated as

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (3.2)$$

Here,  $y_i$  is the observed translation efficiency for sample  $i$ ,  $\hat{y}_i$  is the predicted value,  $\bar{y}$  is the average of all observed values, and  $n$  is the total number of samples.

An ( $R^2$ ) value of 1 means the predictions match the observations exactly. A value of 0 means the model does no better than just using the average of the target variable. Negative values show the model performs worse than this baseline.

### 3.5.3 Spearman correlation coefficient

The Spearman correlation coefficient assesses how well the rankings of the predicted and observed values align [20]. Unlike Pearson correlation [20], it uses the order of the values and does not assume a linear relationship.

Spearman's correlation is often used in bioinformatics because it indicates whether the model and experiments order the samples similarly. Values range from  $-1$  to  $1$ , and higher values mean stronger agreement in rankings.

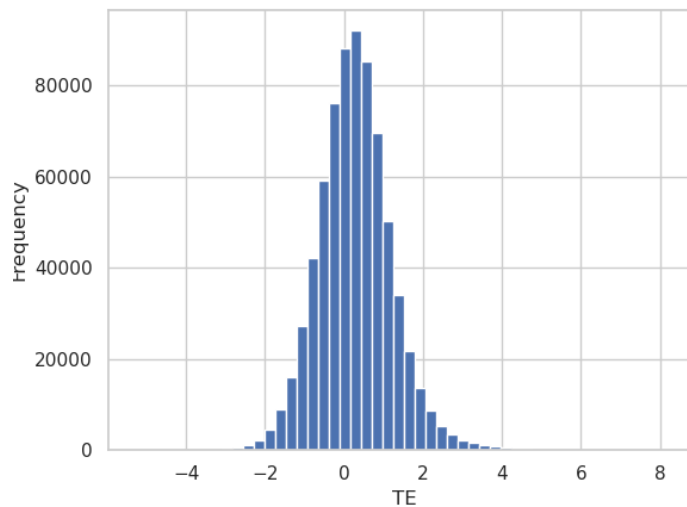
Together, these metrics provide a comprehensive view of model performance. They were used to compare models trained on handcrafted features, models trained on RiNALMo embeddings, and the fine-tuned RiNALMo model.

## 4 Results and discussion

### 4.1 Exploratory Data Analysis

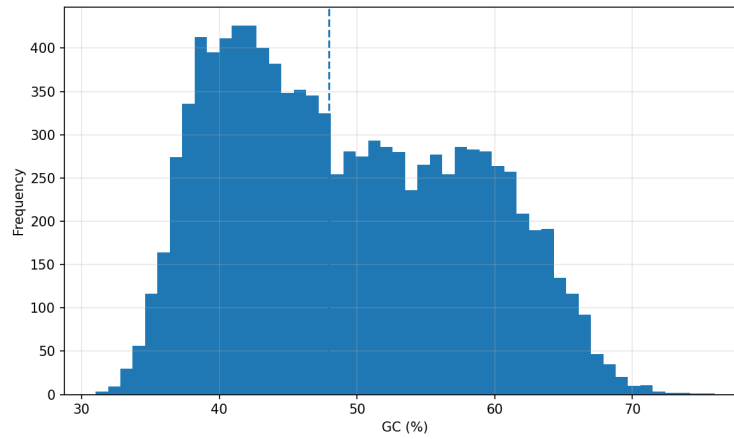
This section analyzes the handcrafted features extracted from the dataset, including sequence composition, codon usage, and transcript structure characteristics, and investigates their potential relevance for translation efficiency prediction.

#### 4.1.1 Translation efficiency distribution

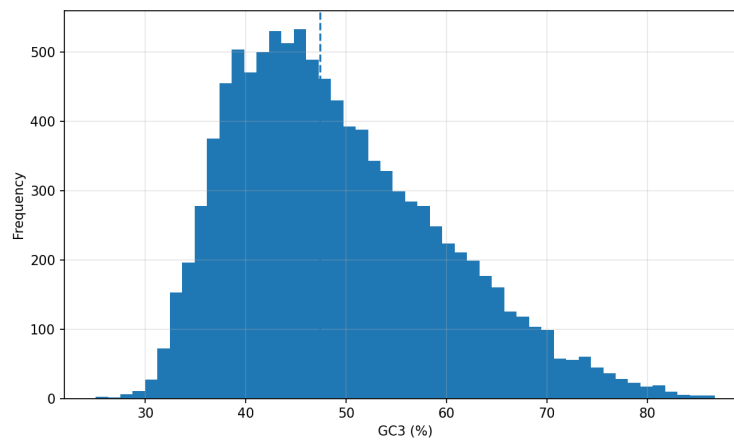


**Figure 4.1:** Distribution of mean translation efficiency values in the dataset.

Figure 4.1 shows the distribution of mean translation efficiency values in the dataset. The distribution is approximately bell-shaped and centered near zero. Most observations are concentrated within a relatively narrow range, while only a small number of transcripts appear in the tails of the distribution.



**Figure 4.2:** Distribution of GC content values.



**Figure 4.3:** Distribution of GC3 content values.

### 4.1.2 GC and GC3 content

Figures 4.2 and 4.3 present the distributions of GC and GC3 content from coding sequences. Both measures vary widely between transcripts.

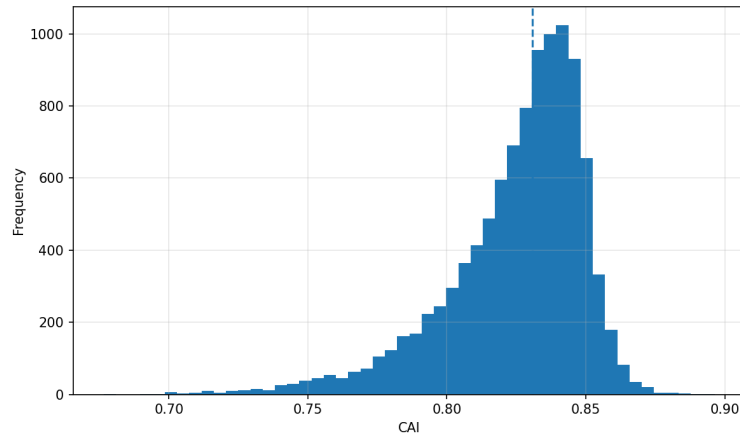
GC content ranges from 32.57% to 79.85%, with a median of 51.70%. GC3 content covers an even wider range, from 22.08% to 98.80%, and has a median of 57.83%.

GC3 content is more spread out than overall GC content, which means there is more variation in nucleotide composition at the third codon position. This suggests that codon usage preferences vary widely across transcripts.

### 4.1.3 Codon adaptation index

Figure 4.4 shows how Codon Adaptation Index (CAI) values are distributed across coding sequences. Most transcripts have high CAI values, with most falling between 0.75





**Figure 4.4:** Distribution of CAI values.

Table 4.1: Summary statistics of CAI values.

| Min   | Median | Max   |
|-------|--------|-------|
| 0.646 | 0.798  | 0.941 |

and 0.85.

The distribution has a clear peak near the median value of 0.798. Most transcripts exhibit CAI values close to this region, while values at the lower and upper ends of the distribution occur less frequently.

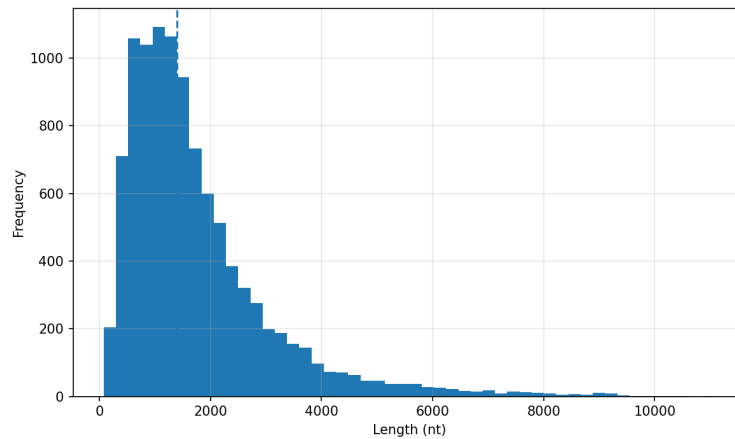
Overall, most coding sequences exhibit similar CAI values, suggesting comparable levels of codon adaptation across the dataset. However, a subset of transcripts exhibits lower CAI values, indicating weaker adaptation to the dataset's preferred codon usage patterns.

#### 4.1.4 Sequence length features

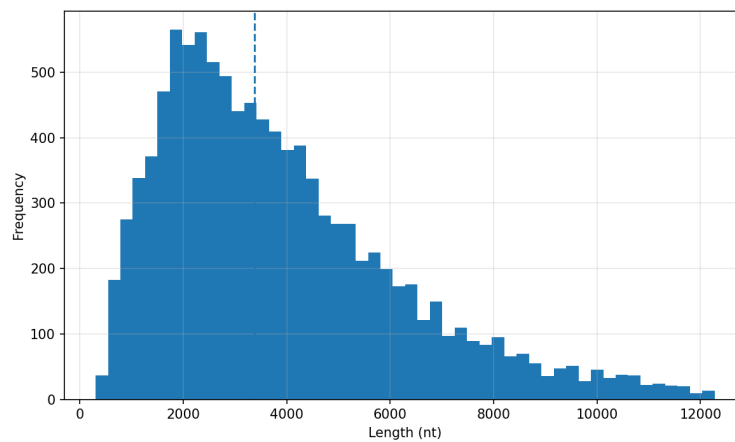
Table 4.2: Summary statistics of sequence length features.

| Feature           | Min | Median | Max   |
|-------------------|-----|--------|-------|
| CDS length        | 78  | 1401   | 11088 |
| Transcript length | 303 | 3372   | 12275 |
| 3'UTR length      | 1   | 1309   | 10968 |
| 5'UTR length      | 20  | 166    | 5140  |

Figures 4.5 and 4.6 show the distributions of CDS and transcript lengths in the dataset. Most transcripts are relatively short, but a few sequences are much longer, which creates a strong right-skew in both distributions.



**Figure 4.5:** Distribution of CDS lengths.



**Figure 4.6:** Distribution of transcript lengths.

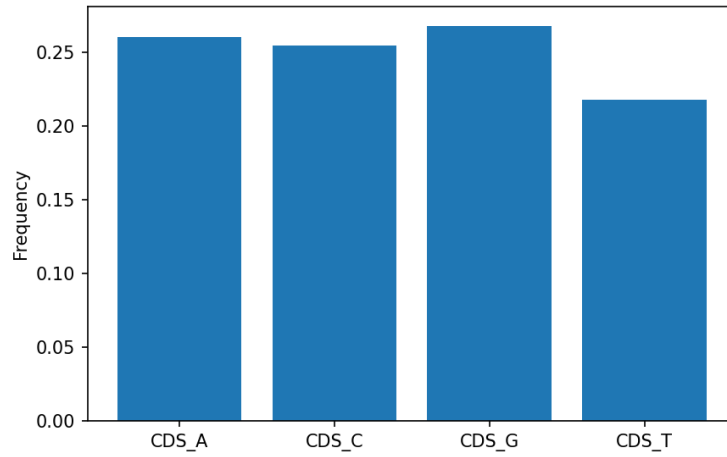
The median CDS length is 1401 nucleotides, while the median transcript length is 3372 nucleotides. This difference comes from the untranslated regions (UTRs) that add to the total transcript length.

While most transcripts are of moderate length, a small number of considerably longer transcripts are also present. This substantial variation in transcript and coding sequence lengths supports their inclusion as potentially informative features for predicting translation efficiency.

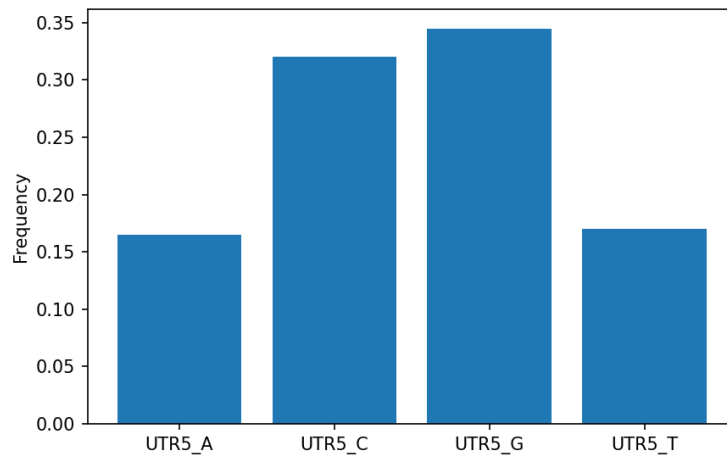
### 4.1.5 Nucleotide composition

Figures 4.7, 4.8, and 4.9 show the average nucleotide composition in coding sequences and untranslated regions.

The CDS regions have a fairly balanced nucleotide composition. Guanine is the most



**Figure 4.7:** Average nucleotide composition in CDS regions.



**Figure 4.8:** Average nucleotide composition in 5'UTR regions.

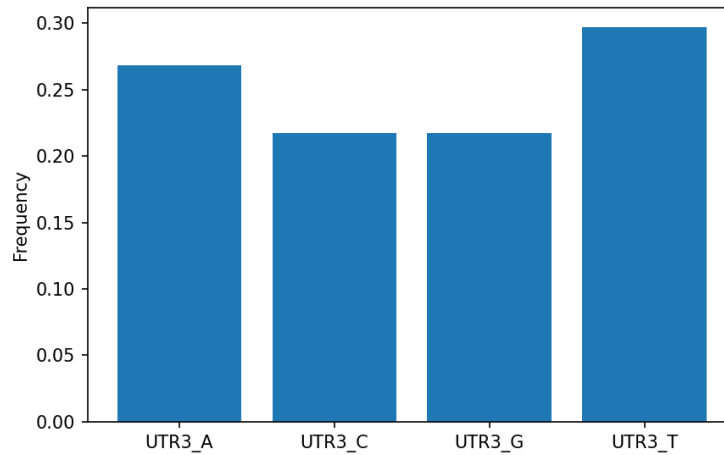
common nucleotide (26.8%), while thymine is the least common (21.8%). In general, the four nucleotides appear in similar amounts in coding sequences.

The untranslated regions have different composition patterns. The 5'UTR contains more guanine and cytosine, making up over 66% of the nucleotides there. In contrast, the 3'UTR has more adenine and uracil, with thymine being the most common at 29.7%.

These results show that nucleotide composition varies a lot between transcript regions. This suggests that the unique sequence features of each region could help predict

Table 4.3: Average nucleotide composition across transcript regions.

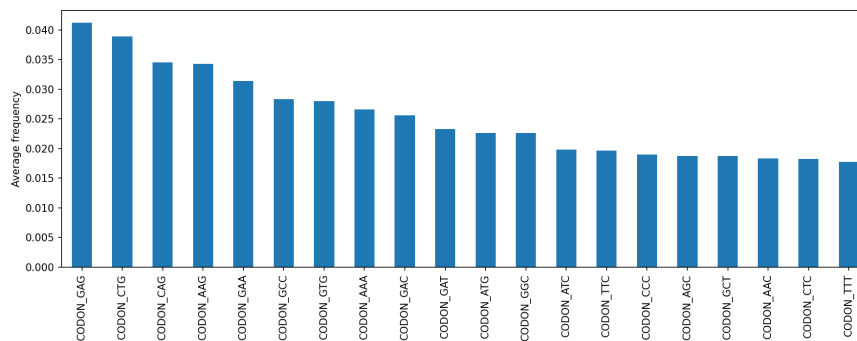
| Region | A     | C     | G     | T     |
|--------|-------|-------|-------|-------|
| CDS    | 0.260 | 0.254 | 0.268 | 0.218 |
| 5'UTR  | 0.165 | 0.320 | 0.345 | 0.170 |
| 3'UTR  | 0.269 | 0.217 | 0.217 | 0.297 |



**Figure 4.9:** Average nucleotide composition in 3'UTR regions.

translation efficiency.

#### 4.1.6 Codon usage



**Figure 4.10:** Twenty most frequent codons observed in the coding sequences.

Figure 4.10 shows the twenty most common codons found in the coding sequences of the dataset. GAG, CTG, CAG, and AAG are the most frequent, accounting for a large share of total codon usage.

Many of the most frequent codons contain guanine and cytosine, consistent with the high GC and GC3 contents observed in the dataset. The codon frequencies are not evenly distributed, indicating codon usage bias across transcripts.

The differences in codon frequencies suggest that codon usage patterns vary between coding sequences. These patterns may help predict translation efficiency.

### 4.1.7 Handcrafted Feature Set

The analyses in this section show that sequence composition, codon usage, and transcript characteristics vary widely across the dataset. Using these findings, we created a set of handcrafted features for later machine learning experiments.

The final feature vector included GC content, GC3 content, Codon Adaptation Index (CAI), coding sequence length, transcript length, codon frequency features, and nucleotide composition features calculated separately for the CDS, 5'UTR, and 3'UTR regions. In total, 81 handcrafted features were extracted for each transcript.

## 4.2 Predictive machine-learning models

To assess the effect of different sequence representations, Linear Regression and LightGBM models were evaluated using both handcrafted features and RiNALMo embeddings. In addition, the pretrained RiNALMo model was fine-tuned for the translation efficiency prediction task.

### 4.2.1 Overall comparison of results

Table 4.4 summarizes the performance of all evaluated approaches for translation efficiency prediction. The results suggest that both the type of input representation and the predictive model affect how well translation efficiency can be predicted.

Table 4.4: Performance comparison of different translation efficiency prediction approaches.

| Input representation | Model               | $R^2 \uparrow$ | MSE $\downarrow$ | Spearman $\uparrow$ |
|----------------------|---------------------|----------------|------------------|---------------------|
| Handcrafted features | Linear Regression   | 0.455          | 0.252            | 0.689               |
| Handcrafted features | LightGBM            | 0.504          | 0.230            | 0.712               |
| RiNALMo embeddings   | Linear Regression   | 0.394          | 0.281            | 0.645               |
| RiNALMo embeddings   | LightGBM            | 0.432          | 0.263            | 0.652               |
| RiNALMo fine-tuning  | Fine-tuned RiNALMo  | 0.245          | 0.338            | 0.474               |
| RiboNN features [11] | LightGBM            | 0.584          | 0.193            | 0.765               |
| RiboNN [11]          | Deep neural network | 0.600          | 0.225            | 0.799               |

*Note.* The reported performance metrics for both the RiboNN LightGBM model and the RiboNN deep neural network were calculated on a common test set rather than taken from the RiboNN paper.

Of all the approaches evaluated, the RiboNN deep neural network achieved the highest overall predictive performance, followed by the LightGBM model, which takes fea-

tures proposed in [11]. Among the models developed in this study, LightGBM trained on handcrafted sequence features produced the best results, outperforming both Linear Regression and the models using RiNALMo embeddings. In contrast, the fine-tuned RiNALMo model with a prediction head achieved the weakest performance.

When comparing handcrafted features to pretrained RiNALMo embeddings, we found that handcrafted sequence features gave better results for predicting translation efficiency in our machine learning models. Both Linear Regression and LightGBM performed better with handcrafted features than with RiNALMo embeddings. This may be because RiNALMo was mainly trained on non-coding RNA sequences, while translation efficiency prediction depends on features of protein-coding mRNA transcripts. Also, many transcripts in our study were much longer than those used to train RiNALMo, which could make it harder to transfer knowledge. It is worth mentioning that the original RiNALMo study showed strong results for translation efficiency prediction using 5'UTR-based datasets [8]. So, the lower performance we saw here is probably due to differences in the pretrained model, the downstream task, the dataset, and how the experiments were set up.

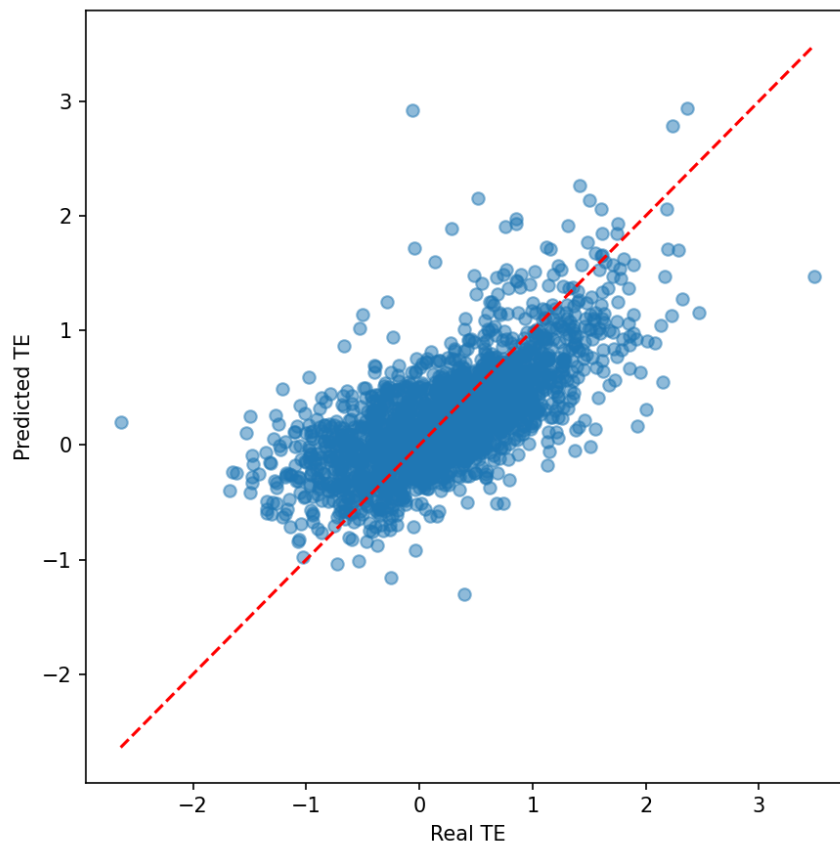
The following sections provide a more detailed analysis of each predictive approach.

#### **4.2.2 Linear regression**

Linear Regression reached an  $R^2$  value of 0.455 with handcrafted features and 0.394 with RiNALMo embeddings. Figures 4.11 and 4.12 illustrate the relationship between the predicted and observed translation efficiency values for both input representations. An  $R^2$  value of 0.455 means that the handcrafted-feature model explains approximately 45.5% of the variability in translation efficiency, representing a moderate fit. This indicates that the selected sequence-derived features capture a substantial part of the biological signal, although a considerable proportion of the variability remains unexplained.

Predictions obtained using handcrafted features are more closely aligned with the diagonal line of perfect agreement than those obtained using RiNALMo embeddings, consistent with the higher  $R^2$  value.

These results point to several possible directions for future research. Because hand-

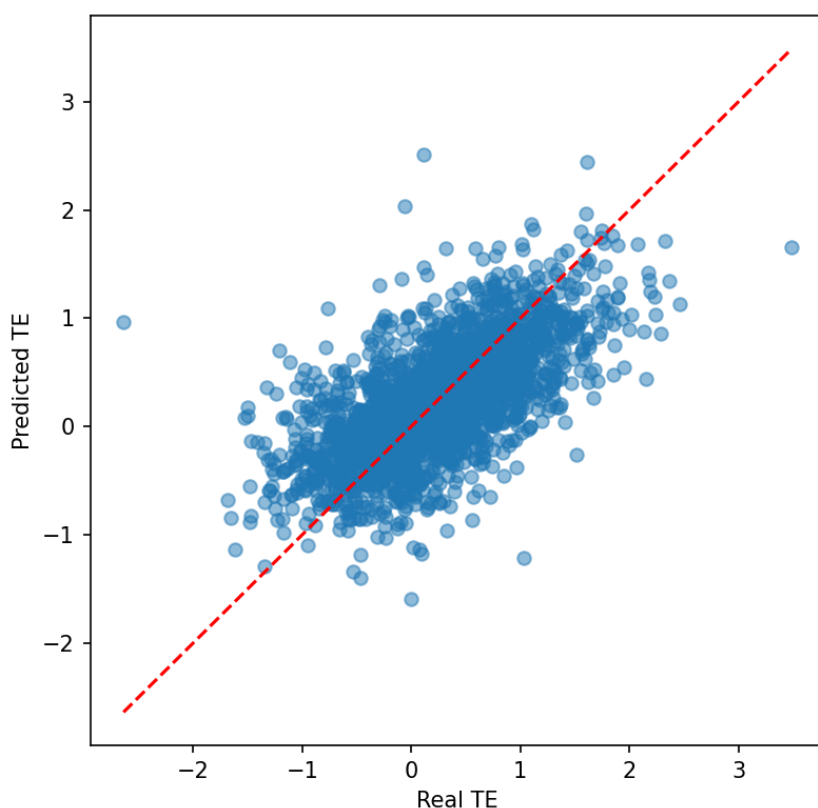


**Figure 4.11:** Predicted versus observed translation efficiency values obtained using Linear Regression trained on handcrafted features

crafted features performed better than RiNALMo embeddings in linear regression, a closer look at how each feature contributes could provoke new biological insights. Although we used LightGBM to explore feature importance, future work could also examine the importance of biological features using linear model coefficients. Results from handcrafted features also suggest that other traditional machine learning methods could be worth exploring. Trying different regression models and ensemble methods may show if further improvements are possible with biologically motivated sequence-derived features.

### 4.2.3 LightGBM

The LightGBM model trained on handcrafted features demonstrated the highest predictive performance in this study, achieving an  $R^2$  of 0.504, an MSE of 0.230, and a Spearman correlation of 0.712. In comparison, the model utilizing RiNALMo embeddings exhibited lower performance, with an  $R^2$  of 0.432, an MSE of 0.263, and a Spearman correlation



**Figure 4.12:** Predicted versus observed translation efficiency values obtained using Linear Regression trained on RiNALMo embeddings.

of 0.652.

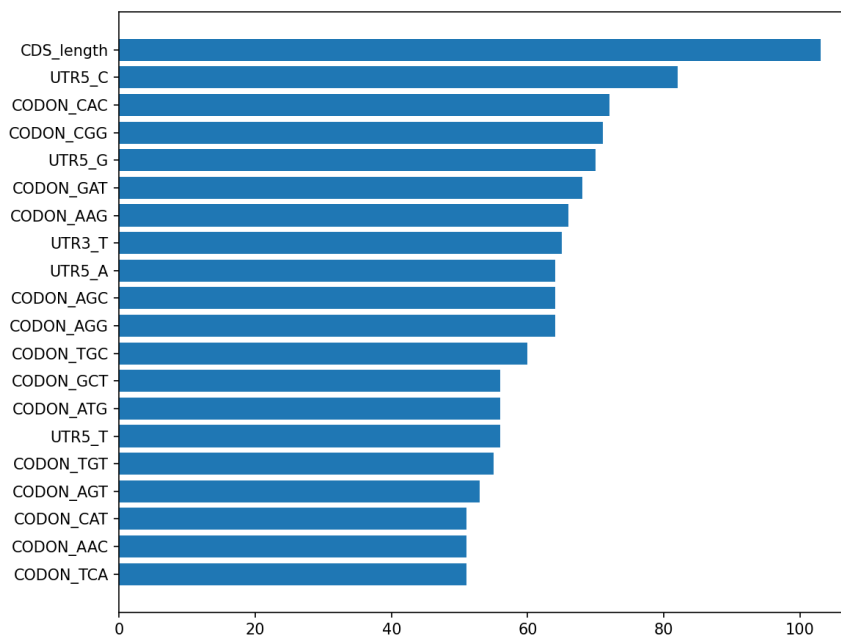
Even though the LightGBM model with handcrafted features performed best among the models developed in this study, its performance remained below that of both the RiboNN LightGBM model and the RiboNN deep neural network evaluated on the same test set. One possible explanation is the more comprehensive feature set used in the RiboNN study, which includes 161 handcrafted sequence-derived features, such as amino acid composition, k-mer frequencies, Kozak sequence characteristics, and RNA secondary-structure features. In comparison, this study used a smaller set of 81 biologically motivated sequence-derived features, most of which represent a subset of those considered in the RiboNN study. The additional sequence descriptors included in RiboNN may therefore provide complementary information that contributes to its higher predictive performance.

Another important difference is the prediction task itself. In the original study, the deep neural network was trained as a multi-task model that simultaneously predicts translation efficiency for 78 human tissues and cell types, from which the mean trans-



lation efficiency can subsequently be calculated. In contrast, this study focused directly on predicting the mean translation efficiency. Predicting tissue-specific translation efficiency may allow the model to capture biological patterns that are averaged out when only the mean value is considered.

Future research could look into predicting translation efficiency for specific tissues, rather than just focusing on the average. This might help reveal how regulation differs between tissues and make it easier to compare with the multi-task prediction strategy from the original study.



**Figure 4.13:** Top 20 most important handcrafted features identified by the LightGBM model for translation efficiency prediction.

To find out which sequence characteristics influenced the predictions most, feature importance scores were taken from the trained LightGBM model, as shown in Figure 4.13. CDS length was the top feature, followed by several nucleotide composition features from the untranslated regions and different codon frequency features.

CDS length was identified as the most important predictor. This suggests that transcript length captures information relevant to predicting translation efficiency. One possible explanation is that longer coding sequences can accommodate more simultaneously translating ribosomes, which may influence the observed translation efficiency values.

Several nucleotide composition features from the 5'UTR, especially cytosine and guanine content, were also among the top predictors. Previous studies have shown that GC-rich 5'UTR regions are associated with increased RNA secondary structure, which can influence translation initiation and translational efficiency [21]. The high importance of cytosine and guanine content indicates that nucleotide composition of the 5'UTR is informative for predicting translation efficiency. This further motivates the inclusion of secondary-structure features, such as minimum free energy.

Interestingly, all four nucleotide composition features derived from the 5'UTR were ranked among the 20 most important predictors. Together with the high importance of GC-related features, this further supports the idea that the sequence composition and structural properties of the 5'UTR contribute substantially to translation efficiency prediction. Future work could investigate whether models based solely on 5'UTR-derived features can achieve competitive predictive performance.

UTR3\_T was also one of the most important features. This matches the enrichment of uracil nucleotides seen in the 3'UTR during exploratory analysis. Further investigation would be required to determine why this feature was ranked so highly and whether it reflects specific mechanisms related to translation regulation.

In summary, the feature importance analysis shows that transcript structure, untranslated region composition, and codon usage all play a major role in predicting translation efficiency. Future studies could focus on these top features to better understand their biological roles in translation efficiency.

#### **4.2.4 RiNALMo fine-tuning**

The fine-tuned RiNALMo model reached an  $R^2$  of 0.245 and an MSE of 0.338 on the test set. Its predictive performance was lower than both the handcrafted feature-based and embedding-based machine learning models. These results indicate that, with the training conditions used in this study, fine-tuning the pretrained RiNALMo model did not fully use the information learned during pretraining for translation efficiency prediction.

Several factors may have contributed to this result. Due to computational constraints,

transcripts longer than 4000 nucleotides were excluded from the training set. In addition, only the prediction head together with the final layers of the pretrained encoder were fine-tuned. Although this is a commonly used approach, other fine-tuning strategies, such as unfreezing additional Transformer blocks or changing the point at which layers are unfrozen during training, were not explored. Furthermore, fine-tuning large pretrained language models typically requires careful hyperparameter optimization and substantial computational resources, limiting the number of configurations that could be evaluated in this study.

Because of these limits, these results should not be seen as a final judgment of RiNALMo’s potential for translation efficiency prediction. Future studies should try more thorough fine-tuning, use larger training sets, and run longer optimization schedules to better test what pretrained RNA language models can do for this task.

## 5 Conclusion

This thesis examined how accurately translation efficiency can be predicted from mRNA sequence data by comparing traditional biologically motivated sequence features with two applications of the pretrained RiNALMo RNA language model: fixed sequence embeddings and task-specific fine-tuning.

The study used the translation efficiency dataset published by Zheng et al. [11] and extracted handcrafted features describing sequence composition, codon usage, and transcript structure. These features were used to train Linear Regression and LightGBM models. In parallel, fixed RiNALMo embeddings were evaluated as an alternative sequence representation, and the pretrained RiNALMo model was further fine-tuned for translation efficiency prediction.

Handcrafted sequence-derived features performed better than RiNALMo embeddings in both machine learning models. With handcrafted features, Linear Regression reached an  $R^2$  of 0.455, while RiNALMo embeddings gave 0.394. LightGBM achieved an  $R^2$  of 0.504 and a Spearman correlation of 0.712 using handcrafted features, compared to 0.432 and 0.652 with RiNALMo embeddings. These results show that biologically motivated sequence descriptors help predict translation efficiency. The feature importance analysis identified CDS length, 5'UTR nucleotide composition, and codon usage as the most informative predictors of translation efficiency.

The fine-tuned RiNALMo model achieved an  $R^2$  of 0.245, which was much lower than the feature-based methods. Still, this result needs to be understood in light of the experimental setup. RiNALMo was mainly pretrained on non-coding RNA sequences and was first tested for translation efficiency prediction using different 5'UTR-based datasets. This study, however, aimed to predict translation efficiency from mRNA transcripts,

which may have made it harder for the model to transfer its knowledge to this new task. Also, due to computational constraints, transcripts longer than 4000 nucleotides were excluded from the training set. Because of these factors, these results should not be seen as a final judgment on the potential of pretrained RNA language models for predicting translation efficiency.

Future studies could look more closely at the biological features highlighted in this research by adding extra sequence-based descriptors, like RNA secondary-structure features, and by examining the 5'UTR in more detail. Researchers might also improve RNA language models by trying more fine-tuning methods, such as unfreezing more Transformer layers, increasing batch sizes, and training for longer periods. Lastly, future work could investigate other RNA language models, particularly those pretrained on mRNA transcripts rather than predominantly non-coding RNAs. Comparing models such as Orthrus [22] or mRNA-LM [23] with RiNALMo could provide further insight into how pretraining data influences translation efficiency prediction.

This study shows that biologically motivated sequence-derived features are still very effective for predicting translation efficiency and offer a solid starting point for classical machine learning models. The results also suggest that pretrained RNA language models could help with translation efficiency prediction, but more work is needed to improve fine-tuning and training methods to get the most out of them. These findings help us better understand how different sequence representations affect translation efficiency prediction and lay the groundwork for future research that combines biological knowledge with deep learning.

## References

- [1] N. Sonenberg and A. G. Hinnebusch, “Regulation of translation initiation in eukaryotes: mechanisms and biological targets,” *Cell*, 2009.
- [2] R. J. Jackson, C. U. Hellen, and T. V. Pestova, “The mechanism of eukaryotic translation initiation and principles of its regulation,” *Nature Reviews Molecular Cell Biology*, 2010.
- [3] H. Gingold and Y. Pilpel, “Determinants of translation efficiency and accuracy,” *Molecular Systems Biology*, 2011.
- [4] A. G. Hinnebusch, I. P. Ivanov, and N. Sonenberg, “Translational control by 5'-untranslated regions of eukaryotic mRNAs,” *Science*, 2016.
- [5] U. Sahin, K. Karikó, and Ö. Türeci, “mRNA-based therapeutics—developing a new class of drugs,” *Nature reviews Drug discovery*, vol. 13, no. 10, pp. 759–780, 2014.
- [6] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. u. Kaiser, and I. Polosukhin, “Attention is All you Need,” in *Advances in Neural Information Processing Systems*, I. Guyon, U. V. Luxburg, S. Bengio, H. Wallach, R. Fergus, S. Vishwanathan, and R. Garnett, Eds., vol. 30. Curran Associates, Inc., 2017. [Online]. Available: [https://proceedings.neurips.cc/paper\\_files/paper/2017/file/3f5ee243547dee91fbd053c1c4a845aa-Paper.pdf](https://proceedings.neurips.cc/paper_files/paper/2017/file/3f5ee243547dee91fbd053c1c4a845aa-Paper.pdf)
- [7] J. Chen, Z. Hu, S. Sun, Q. Tan, Y. Wang, Q. Yu, L. Zong, L. Hong, J. Xiao, T. Shen *et al.*, “Interpretable RNA foundation model from unannotated data for highly accurate RNA structure and function predictions,” *arXiv preprint arXiv:2204.00300*, 2022.

- [8] R. J. Penić, T. Vlašić, R. G. Huber, Y. Wan, and M. Šikić, “RiNALMo: General-Purpose RNA Language Models Can Generalize Well on Structure Prediction Tasks,” *Nature Communications*, 2024.
- [9] Y. Ji, Z. Zhou, H. Liu, and R. V. Davuluri, “DNABERT: pre-trained Bidirectional Encoder Representations from Transformers model for DNA-language in genome,” *Bioinformatics*, vol. 37, no. 15, pp. 2112–2120, 2021.
- [10] N. Brandes, D. Ofer, Y. Peleg, N. Rappoport, and M. Linial, “ProteinBERT: a universal deep-learning model of protein sequence and function,” *Bioinformatics*, vol. 38, no. 8, pp. 2102–2110, 2022.
- [11] D. Zheng, L. Persyn, J. Wang, Y. Liu, F. Ulloa-Montoya, C. Cenik, and V. Agarwal, “Predicting the translation efficiency of messenger RNA in mammalian cells,” *Nature Biotechnology*, 2025.
- [12] P. M. Sharp and W.-H. Li, “The Codon Adaptation Index—A measure of directional synonymous codon usage bias, and its potential applications,” *Nucleic Acids Research*, 1987.
- [13] Y. Liu, S. Rao, I. Hoskins, M. Geng, Q. Zhao, J. Chacko, V. Ghatpande, K. Qi, L. Persyn, J. Wang *et al.*, “Translation efficiency covariation across cell types is a conserved organizing principle of mammalian transcriptomes,” *bioRxiv*, pp. 2024–08, 2025.
- [14] K. Leppeck, R. Das, and M. Barna, “Functional 5’ UTR mRNA structures in eukaryotic translation regulation and how to find them,” *Nature Reviews Molecular Cell Biology*, 2018.
- [15] F. Hia, S. F. Yang, Y. Shichino, M. Yoshinaga, Y. Murakawa, A. Vandenbon, A. Fukao, T. Fujiwara, C. Ono, J. Qu *et al.*, “The effects of codon bias and optimality on mRNA and protein expression,” *Molecular Systems Biology*, 2020.
- [16] G. Ke, Q. Meng, T. Finley, T. Wang, W. Chen, W. Ma, Q. Ye, and T.-Y. Liu, “LightGBM: A highly efficient gradient boosting decision tree,” *Advances in neural information processing systems*, vol. 30, 2017.

- [17] Y. Chu, D. Yu, Y. Li, K. Huang, Y. Shen, L. Cong, J. Zhang, and M. Wang, “A 5’ UTR language model for decoding untranslated regions of mRNA and function predictions,” *Nature Machine Intelligence*, vol. 6, no. 4, pp. 449–460, 2024.
- [18] M. Hossin and M. N. Sulaiman, “A review on evaluation metrics for data classification evaluations,” *International journal of data mining & knowledge management process*, vol. 5, no. 2, p. 1, 2015.
- [19] G. J. Hahn, “The coefficient of determination exposed,” *Chemtech*, vol. 3, no. 10, pp. 609–612, 1973.
- [20] M. Shaqiri, T. Iljazi, L. Kamberi, and R. Ramani-Halili, “Differences between the correlation coefficients Pearson, Kendall and Spearman,” *Journal of Natural Sciences and Mathematics of UT*, vol. 8, no. 15-16, pp. 392–397, 2023.
- [21] M. Kozak, “An analysis of 5’-noncoding sequences from 699 vertebrate messenger RNAs,” *Nucleic acids research*, vol. 15, no. 20, pp. 8125–8148, 1987.
- [22] P. Fradkin, R. hi, T. Dalal, K. Isaev, B. J. Frey, L. J. Lee, Q. Morris, and B. Wang, “Orthrur: toward evolutionary and functional RNA foundation models,” *Nature Methods*, pp. 1–11, 2026.
- [23] S. Li, S. Noroozizadeh, S. Moayedpour, L. Kogler-Anele, Z. Xue, D. Zheng, F. U. Montoya, V. Agarwal, Z. Bar-Joseph, and S. Jager, “mRNA-LM: full-length integrated SLM for mRNA analysis,” *Nucleic Acids Research*, vol. 53, no. 3, p. gkaf044, 2025.



## Statement on the Use of Artificial Intelligence

The following artificial intelligence tools were used during the preparation of this thesis. ChatGPT (OpenAI, GPT - 5.5 version) and Claude (Anthropic) were used to support language-related tasks, including stylistic improvements, grammar checking, and translation of parts of the text. Grammarly was used for grammar and spelling correction, as well as improving writing clarity. These tools were used only to assist with language editing and did not contribute to the scientific content, experimental results, analysis, or conclusions of this thesis.

**Reflections on the Use of Artificial Intelligence:** Artificial intelligence supported the preparation of this thesis. These tools helped with style, grammar, and translating some sections, which made the writing clearer and the process more efficient.

All AI-generated suggestions were carefully reviewed, revised, and adapted before being included in the final thesis. Although these tools helped improve the efficiency of language editing, they occasionally produced inaccurate translations or stylistic suggestions that did not fit the intended context. Therefore, all written content was carefully reviewed before being incorporated into the final version of the thesis.

**Statement on the Use of Artificial Intelligence:** I hereby confirm that artificial intelligence was used in accordance with the Policy on the Appropriate Use of Artificial Intelligence at the Faculty of Electrical Engineering and Computing during the preparation of this Bachelor's thesis, and that the artificial intelligence did not substitute my critical thinking or my original contribution.